

Maximum Likelihood

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Hansen: Probability and Statistics

Chap. 10

Distribution and Density Functions

- $X \sim F$, $F(x) = P(X \leq x)$ is the cumulative dist. function.
- Discrete random variables:
 - the probability mass function: $\pi(x) = P(X = x)$
 - the distribution at the support points τ_j is
$$F(\tau_j) = \sum_{k=1}^j \pi(\tau_k),$$
- If $F(x)$ is continuous, X is a continuous r.v.
 - If $F(x)$ is differentiable, $f(x) = \frac{d}{dx} F(x)$, is the probability density function (pdf).
 - $f(x) \geq 0$ for all x , and $\int_{-\infty}^{\infty} f(x) dx = 1$
 - For given α , quantile $q(\alpha)$ is such that $\alpha = F(q(\alpha))$.

- A parametric model for X is a probability function $f(X|\theta)$ that depends on an unknown parameter vector $\theta \in \Theta$. The set Θ is called the parameter space.
- A model for a random sample $X_i, i = 1, \dots, n$ assumes that every X_i has density $f(x|\theta)$
- A parametric model specifies that the population distribution $f(x|\theta)$ belongs to a specific parametric family.
 - X normally distributed, or $X \sim N(\mu, \sigma^2)$:

$$f(x|\mu, \sigma^2) = \sigma^{-1} \phi\left(\frac{(x - \mu)}{\sigma}\right)$$

- X is exponentially distributed with parameter $\lambda > 0$

$$f(x|\lambda) = \frac{1}{\lambda} \exp(-x/\lambda).$$

- A model is correctly specified if there is a unique $\theta_0 \in \Theta$ s.t. $f(x|\theta_0) = f(x)$, the true distribution.

Likelihood

- The joint density of the observations calculated using the model parameterized by θ is $f(x_1, \dots, x_n | \theta)$.
- The likelihood $L_n(\theta)$ is the joint density evaluated at the observed data, and viewed as a function of θ .
- For independent observations with identical densities,

$$L_n(\theta) = f(x_1, \dots, x_n | \theta) = f(x_1 | \theta) f(x_2 | \theta) \dots f(x_n | \theta) = \prod_{i=1}^n f(x_i | \theta).$$

- the log-likelihood function is

$$\ell_n(\theta) = \log L_n(\theta) = \sum_{i=1}^n \log f(x_i | \theta)$$

	probability distribution	likelihood function
	$f(x \theta)$	$L(\theta)$
what is fixed	parameters (eg μ, λ, p)	observed data x
what varies	data	parameters
question	given parameters, plausibility of specific data	given data, plausibility of parameters
Integration	$\int_{-\infty}^{\infty} f(x \theta)dx = 1$	$\int_{-\infty}^{\infty} L(\theta)d\theta \neq 1$
Derivative	w.r.t. x	w.r.t. θ

Because the likelihood function does not integrate to 1, we don't say probability of θ when we look at a likelihood.

Maximum likelihood estimator:

$$\hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} \ell_n(\theta).$$

Examples:

- exponential distribution (eg. waiting time)

$f(x|\lambda) = \lambda^{-1} \exp(-x/\lambda)$. Then $\hat{\lambda} = \bar{x}_n$.

$$\text{FOC} \quad 0 = \frac{d\ell_n(\lambda)}{d\lambda} = -\frac{n}{\lambda} + n\frac{\bar{x}_n}{\lambda^2}.$$

- Bernoulli distribution (eg. coin flip):

$f(x|p) = p^x(1-p)^{1-x}$. Then $\hat{p} = \bar{x}_n$ since

$$\text{FOC} \quad 0 = \frac{d\ell_n(p)}{dp} = \frac{n\bar{x}_n}{p} - \frac{n(1-\bar{x}_n)}{1-p}.$$

To find MLE

- i specify $f(x|\theta)$ as a function of x and θ
- ii take log to get $\log f(x|\theta)$
- iii evaluate likelihood at $x = x_i$ to get $\ell_n(\theta) = \sum_{i=1}^n \log f(x_i|\theta)$.
- iv solve FOC to get $\hat{\theta}$
- v check second order condition

Gaussian MLE: $\theta = \mu$

- For $X \sim N(\mu, \sigma_0^2)$, the density of X with σ^2 known is

$$f(x|\mu) = \frac{1}{(2\pi\sigma_0^2)^{1/2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma_0^2}\right)$$

- the log density is

$$\log f(x|\mu) = -\frac{1}{2} \log(2\pi\sigma_0^2) - \frac{(x-\mu)^2}{2\sigma_0^2}.$$

- FOC for $\hat{\mu}$: gives $\hat{\mu}_n = \bar{x}_n$

$$\frac{\partial}{\partial \mu} \ell_n(\hat{\mu}) = \frac{1}{\sigma_0^2} \sum_{i=1}^n (x_i - \hat{\mu}) = 0$$

- SOC: $\frac{\partial^2}{\partial \mu^2} \ell_n(\hat{\mu}) = -\frac{n}{\sigma_0^2} < 0$ as required.

Gaussian MLE: $\theta = \sigma^2$

- For $X \sim N(\mu_0, \sigma^2)$, the density of X with μ known is

$$f(x|\sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left(-\frac{(x-\mu_0)^2}{2\sigma^2}\right)$$

- log density: $\log f(x|\sigma^2) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu_0)^2}{2\sigma^2}$.
- log likelihood: $\ell_n(\sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{\sum_i (x_i - \mu_0)^2}{2\sigma^2}$.
- $\hat{\sigma}^2 = \frac{1}{n} \sum_i (x_i - \mu_0)^2$ solves FOC $0 = -\frac{n}{2\hat{\sigma}^2} + \frac{\sum_i (x_i - \mu_0)^2}{2\hat{\sigma}^4}$.
- second order condition is $\frac{\partial^2}{\partial \sigma^2} \ell_n(\hat{\sigma}^2) = -\frac{n}{2\hat{\sigma}^4} < 0$.

Likelihood Analog Principle

- Define expected log density $\ell(\theta) = E[\log f(x|\theta)]$. Then

$$\theta_0 = \operatorname{argmax}_{\theta} \ell(\theta).$$

- Sample average likelihood: $\bar{\ell}_n(\theta) = \frac{1}{n} \ell_n(\theta) = \frac{1}{n} \sum_i \log f(x_i|\theta)$.

$$\hat{\theta} = \operatorname{argmax}_{\theta} \bar{\ell}_n(\theta).$$

Invariance property: If $\hat{\theta}$ is MLE of θ , any transformation $\hat{\beta} = h(\hat{\theta})$ is MLE of $\beta = h(\theta)$.

Score Vector and Fisher Information

- the score vector of θ is a vector of first partial derivatives:

$$S_n(\theta) = \frac{\partial}{\partial \theta} \ell_n(\theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f(x_i | \theta).$$

It tells us how sensitive is the likelihood to the parameter vector. If $\hat{\theta}$ is an interior solution, $S_n(\hat{\theta}) = 0$.

- the derivative of the log-likelihood for a single observation is $s_i = \frac{\partial}{\partial \theta} \log f(x_i | \theta)$.

Hessian Matrix

- the Hessian is the second derivative matrix of the likelihood

$$H_n(\theta) = \frac{\partial^2}{\partial \theta \partial \theta'} \ell_n(\theta) = \sum_{i=1}^n \frac{\partial^2}{\partial \theta \partial \theta'} \log f(x_i | \theta).$$

It indicates the curvature of the log likelihood. Smaller values indicate a flatter likelihood.

- the expected Hessian evaluated at θ_0 is

$$\mathcal{H}_\theta = \frac{\partial^2}{\partial \theta \partial \theta'} \ell(\theta_0) = E \left[\frac{\partial^2}{\partial \theta \partial \theta'} \log f(x | \theta_0) \right].$$

Cramer Rao Lower Bound

- If the model is correctly specified, expected efficient score is

$$E[s_i^0] = E\left[\frac{\partial}{\partial \theta} \log f(x_i|\theta_0)\right] = \frac{\partial}{\partial \theta} E[\log f(x_i|\theta_0)] = \frac{\partial}{\partial \theta} \ell(\theta_0) = 0$$

- Fisher Information is the variance of the efficient scores:

$$\mathcal{I}_\theta = E[s_i^0 s_i^{0'}].$$

- Information matrix equality: if the model is **correctly specified** and the support of X does not depend on θ ,

$$\mathcal{I}_\theta = -\mathcal{H}_\theta.$$

- CRLB: If $\tilde{\theta}$ is an unbiased estimator of θ , $\text{var}(\tilde{\theta}) \geq (n\mathcal{I}_\theta)^{-1}$.

An unbiased vector $\tilde{\theta}$ must have a covariance matrix that is bounded from below by the inverse of Fisher information matrix. scaled by the sample size.

Proof of Information Matrix Equality

$$\begin{aligned}-\mathcal{H}_\theta &= -\frac{\partial^2}{\partial\theta\partial\theta'} E[\log f(X|\theta)] \\&= -\frac{\partial}{\partial\theta} E\left[\frac{\frac{\partial}{\partial\theta'} f(X|\theta)}{f(X|\theta)}\right] + E\left[\frac{\frac{\partial}{\partial\theta} f(X|\theta) \frac{\partial}{\partial\theta'} f(X|\theta)}{f(X|\theta)^2}\right] \\&= A + \mathcal{I}(\theta)\end{aligned}$$

But $A = 0$ at $\theta = \theta_0$ since

$$\begin{aligned}-\frac{\partial^2}{\partial\theta\partial\theta'} E\left[\frac{f(X|\theta)}{f(X|\theta_0)}\right]_{\theta=\theta_0} &= -\frac{\partial^2}{\partial\theta\partial\theta'} \int \frac{f(X|\theta)}{f(X|\theta_0)} f(X|\theta_0) dx|_{\theta=\theta_0} \\&= -\frac{\partial^2}{\partial\theta\partial\theta'} \underbrace{\int f(X|\theta) dx|_{\theta=\theta_0}}_{=1} = 0.\end{aligned}$$

Thus $\mathcal{I}(\theta) = -\mathcal{H}(\theta)$.

Example 1: $X \sim N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$

- log density: $\log f(x|\mu, \sigma^2) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \sigma^2 - \frac{(x-\mu)^2}{2\sigma^2}$

- first derivatives

$$\frac{\partial}{\partial \mu} \log f(x|\mu, \sigma^2) = \frac{x - \mu}{\sigma^2}$$

$$\frac{\partial}{\partial \sigma^2} \log f(x|\mu, \sigma^2) = -\frac{1}{2\sigma^2} + \frac{(x - \mu)^2}{\sigma^4}$$

- second derivatives

$$\frac{\partial^2}{\partial \mu^2} \log f(x|\mu, \sigma^2) = -\frac{1}{\sigma^2}$$

$$\frac{\partial^2}{\partial (\sigma^2)^2} \log f(x|\mu, \sigma^2) = \frac{1}{2\sigma^4} - \frac{(x - \mu)^2}{\sigma^6}$$

$$\frac{\partial^2}{\partial \mu \partial \sigma^2} \log f(x|\mu, \sigma^2) = -\frac{(x - \mu)}{\sigma^4}$$

- expected Fisher Information:

$$\mathcal{I}_\theta = E \begin{pmatrix} \frac{1}{\sigma^2} & \frac{X-\mu}{\sigma^4} \\ \frac{X-\mu}{\sigma^4} & \frac{(X-\mu)^2}{\sigma^6} - \frac{1}{2\sigma^4} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{pmatrix}$$

- the CRLB is

$$(n\mathcal{I}_\theta)^{-1} = \begin{pmatrix} \sigma^2/n & 0 \\ 0 & \frac{2\sigma^4}{n} \end{pmatrix}$$

- Example: Let $s^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$. Then $\text{var}(s^2) \geq \frac{2\sigma^4}{n}$.

Example 2 : $X \sim N(0, \sigma^2)$, $\mu = 0$ known

Here, $\theta = \sigma^2$.

- The log density $\log f(x_i|\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log \theta - \frac{x_i^2}{2\theta}$

$$s_i(\theta) = \frac{d}{d\theta} \log f(x_i|\theta) = -\frac{1}{2\theta} + \frac{x_i^2}{2\theta^2} = \frac{x_i^2 - \theta}{2\theta^2},$$

$$H_i(\theta) = \frac{d^2}{d\theta^2} \log f(x_i|\theta) = \frac{1}{2\theta^2} - \frac{x_i^2}{\theta^3} = \frac{\theta - 2x_i^2}{2\theta^3},$$

- $E[\log(f(x_i|\theta))] = \ell(\theta) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\theta) - \frac{\theta_0}{2\theta}$ with

$$E[s_i^0] = E\left[\frac{x_i^2 - \theta_0}{2\theta_0^2}\right] = 0, \quad \text{var}(s_i) = \frac{E[(x_i^2 - \theta_0)^2]}{4\theta_0^4} = \frac{1}{2\theta_0^2} = \mathcal{I}(\theta)$$

- $\mathcal{H}_\theta = E[H_i^0] = -\frac{1}{2\theta_0^2}$. Hence $\mathcal{I}_\theta = -\mathcal{H}_\theta$.

The Normal Linear Model

- $y_i = x_i'\beta + e_i$, $e_i \sim N(0, \sigma^2)$:
- the multivariate normal density for e is

$$f(e) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} e'e\right).$$

The multivariate density for y conditional on x is

$$f(y|X) = f(e) \left| \frac{de}{dy} \right|$$

where $\left| \frac{de}{dy} \right|$ is the absolute value of the determinant of the $n \times n$ matrix of partial derivatives of the elements of e wrt to y . In this linear model, the determinant is I_n . Thus

$$f(y|x) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{\sum_i (y_i - x_i'\beta)^2}{2\sigma^2}\right).$$

- The likelihood is the joint pdf of the data evaluated at the observed sample, viewed as a function of the parameters.
- Assuming that the observations are mutually independent,

$$\begin{aligned}f(y_1, \dots, y_n | x_1, \dots, x_n) &= \prod_{i=1}^n f(y_i | x_i) \\&= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(\frac{-\sum_{i=1}^n (y_i - x_i' \beta)^2}{2\sigma^2}\right) \\&\equiv L_n(\beta, \sigma^2).\end{aligned}$$

MLE for $\theta = (\beta, \sigma^2)$.

The MLE is the value of θ that maximizes the log likelihood.

$$\ell_n(\beta, \sigma^2) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \frac{1}{2} \sum_{i=1}^n \left[\frac{(y_i - x_i' \beta)^2}{\sigma^2} \right]$$

$$S(\theta) = \begin{pmatrix} \frac{\partial \ell_n}{\partial \beta} \\ \frac{\partial \ell_n}{\partial \sigma^2} \end{pmatrix} = \begin{pmatrix} \frac{X'e}{\sigma^2} \\ -\frac{n}{2\sigma^2} + \frac{e'e}{2\sigma^4} \end{pmatrix}$$

$$S(\theta)|_{\theta=\hat{\theta}} = \begin{pmatrix} \frac{1}{\hat{\sigma}_{MLE}^2} \sum_i x_i (y_i - x_i' \hat{\beta}_{MLE}) \\ -\frac{n}{2\hat{\sigma}_{MLE}^2} + \frac{1}{2\hat{\sigma}_{MLE}^4} \sum_i (y_i - x_i' \hat{\beta}_{MLE})^2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Maximizing $\ell_n(\beta, \sigma^2)$ w.r.t. β is the same as $\min \text{SSR}(\beta)$. Thus

$$\begin{aligned}\hat{\beta}_{MLE} &= (X'X)^{-1}X'Y \equiv \hat{\beta}_{OLS} \\ \hat{\sigma}_{MLE}^2 &= \frac{1}{n} \sum_{i=1}^n (y_i - x_i' \hat{\beta}_{MLE})^2.\end{aligned}$$

Since $\frac{1}{2\hat{\sigma}^2} (y - X\hat{\beta})'(y - X\hat{\beta}) = \frac{1}{2} \frac{n\hat{\sigma}^2}{\hat{\sigma}^2} = n/2$,

$$\ell_n(\hat{\beta}_{MLE}, \hat{\sigma}_{MLE}^2) = -\frac{n}{2} \log(2\pi\hat{\sigma}_{MLE}^2) - \frac{n}{2}.$$

Summary:

- $\hat{\beta} = (X'X)^{-1}X'Y$ can be motivated by methods of moments, minimizing mean-squared prediction error, or maximizing the Gaussian likelihood. While MM and OLS do not require distributional assumptions, MLE is fully parametric.

Misspecified Likelihood and KLIC

- Kullback-Leibler divergence between two densities f and g :

$$\text{KLIC}(f, g) = \int f(x) \log \left(\frac{f(x)}{g(x)} \right) dx$$

$$\text{KLIC}(f, f) = 0, \text{KLIC}(f, g) \geq 0, f = \operatorname{argmin}_g \text{KLIC}(f, g).$$

- Recall that a model is correctly specified if the true density is a member of the parametric family. So KLIC is minimized by setting θ to its true value. ie. $\theta_0 = \operatorname{argmin}_{\theta \in \Theta} \text{KLIC}(f, f_\theta)$.
- We would like $f(x) = f(x|\theta_0)$ for some $\theta_0 \in \Theta$ (the parameter space). But what if we didn't? Despite misspecification, a parametric model can still be a good approximation to a given density.

Pseudo-True Model

- Now let θ_0 be the pseudo-true parameter for a model f_θ is the one closest to the true density $\theta_0 = \operatorname{argmin}_{\theta \in \Theta} \text{KLIC}(f, f_\theta)$:
When the model is correctly specified, the pseudo true value is the true value.

$$\begin{aligned}\text{KLIC}(f, f_\theta) &= \underbrace{\int f(x) \log f(x) dx}_{\text{does not depend on } \theta} - \underbrace{\int f(x) \log f(x|\theta) dx}_{E[\log f(X|\theta)] = \ell(\theta)} \\ &= \int f(x) \log f(x) dx - \ell(\theta)\end{aligned}$$

- Theorem: under misspecification, the pseudo true parameter $\theta_0 = \operatorname{argmax} \ell(\theta)$.
- But under misspecification, $\mathcal{I}_\theta \neq \mathcal{H}_\theta$.

QMLE for Omitted Heteroskedasticity

DGP: $y_i = x_i' \beta + e_i$, $e_i \sim N(0, \sigma^2(x_i))$.

- We assume a Gaussian likelihood that assumes $e_i \sim N(0, 1)$,

$$\frac{\partial}{\partial b} \log f(y_i|x; b) = x_i(y_i - x_i b)$$

$$\frac{\partial^2}{\partial b \partial b'} \log f(y_i|x; b) = -x_i x_i'.$$

- We obtain $\hat{\beta} = (X'X)^{-1}X'y$,
- $\hat{\beta}$ is the quasi-MLE. It is consistent, but not efficient. The assumed Gaussian distribution is incorrect, but it belongs to the linear-exponential family.

Deriving the Score

$$\nabla_{\beta} \ell(\beta, \sigma^2) = \frac{\partial \ell}{\partial \beta} = -\frac{1}{2\sigma^2} \sum_{i=1}^n \frac{\partial}{\partial \beta} (y_i - x_i' \beta)^2$$

Since $\frac{\partial}{\partial \beta} (y_i - x_i' \beta)^2 = 2(y_i - x_i' \beta)(-x_i) = -2x_i(y_i - x_i' \beta)$.

$$\begin{aligned} \nabla_{\beta} \ell(\beta, \sigma^2) &= \frac{1}{\sigma^2} \sum_{i=1}^n x_i (y_i - x_i' \beta) \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n x_i e_i \end{aligned}$$

Variance of the Score (The “Meat” $\mathcal{I}$)

We have $\mathbb{E}[e_i | x_i] = 0$ and $\mathbb{E}[e_i^2 | x_i] = \sigma_i^2$. The variance of the score (the outer product of the gradient) is defined as:

$$\begin{aligned}\mathcal{I} &= \text{Var}(\nabla_{\beta} \ell) = \mathbb{E} [(\nabla_{\beta} \ell)(\nabla_{\beta} \ell)'] \\&= \mathbb{E} \left[\left(\frac{1}{\sigma^2} \sum_{i=1}^n x_i e_i \right) \left(\frac{1}{\sigma^2} \sum_{j=1}^n x_j e_j \right)' \right] \\&= \frac{1}{\sigma^4} \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[x_i x_j' e_i e_j] = \frac{1}{\sigma^4} \sum_{i=1}^n \mathbb{E}[x_i x_i' e_i^2] \quad \text{by iid} \\&= \frac{1}{\sigma^4} \sum_{i=1}^n \mathbb{E}[\sigma_i^2 x_i x_i']\end{aligned}$$

where the last line uses

$$\mathbb{E}[x_i x_i' e_i^2] = \mathbb{E}[\mathbb{E}[x_i x_i' e_i^2 | x_i]] = \mathbb{E}[x_i x_i' \mathbb{E}[e_i^2 | x_i]] = \mathbb{E}[\sigma_i^2 x_i x_i']$$

the Expected Negative Hessian (The “Bread” $\mathcal{H}$)

$$H(\beta) = \frac{\partial^2 \ell}{\partial \beta \partial \beta'} = \frac{\partial}{\partial \beta'} \left[\frac{1}{\sigma^2} \sum_{i=1}^n x_i (y_i - x_i' \beta) \right] = -\frac{1}{\sigma^2} \sum_{i=1}^n x_i x_i'$$

since $\frac{\partial}{\partial \beta'} (y_i - x_i' \beta) = -x_i'$. The information matrix component A is

$$\mathcal{H} = \mathbb{E}[-H(\beta)] = \frac{1}{\sigma^2} \sum_{i=1}^n \mathbb{E}[x_i x_i']$$

which depends entirely on the assumed constant variance parameter σ^2 and $\mathbb{E}[x_i x_i']$.

Sandwich Variance, $\text{Asy.Var}(\hat{\beta}) = \mathcal{H}^{-1}\mathcal{I}\mathcal{H}^{-1}$

$$\begin{aligned}\text{Asy.Var}(\hat{\beta}) &= \mathcal{H}^{-1}\mathcal{I}\mathcal{H}^{-1} \\ &= \left(\frac{1}{\sigma^2} \sum_{i=1}^n \mathbb{E}[x_i x_i'] \right)^{-1} \left(\frac{1}{\sigma^4} \sum_{i=1}^n \mathbb{E}[\sigma_i^2 x_i x_i'] \right) \left(\frac{1}{\sigma^2} \sum_{i=1}^n \mathbb{E}[x_i x_i'] \right)^{-1} \\ &= \left(\sum_{i=1}^n \mathbb{E}[x_i x_i'] \right)^{-1} \left(\sum_{i=1}^n \mathbb{E}[\sigma_i^2 x_i x_i'] \right) \left(\sum_{i=1}^n \mathbb{E}[x_i x_i'] \right)^{-1}\end{aligned}$$

since $\left(\frac{1}{\sigma^2}\right)^{-1} \left(\frac{1}{\sigma^4}\right) \left(\frac{1}{\sigma^2}\right)^{-1} = \sigma^2 \cdot \frac{1}{\sigma^4} \cdot \sigma^2 = 1$. Because $\sigma_i^2 \neq \sigma^2$,

$$\mathcal{H} = \frac{1}{\sigma^2} \sum_{i=1}^n \mathbb{E}[x_i x_i'] \quad \neq \quad \mathcal{I} = \frac{1}{\sigma^4} \sum_{i=1}^n \mathbb{E}[\sigma_i^2 x_i x_i']$$